Lecture 1. Arithmetic, fractions, ratios, and percentages

Notation	Meaning and example
Letters such as a, x, n	Variables represent numbers or quantities. Their meanings depend on the problem: n might count repetitions, while x might be an unknown amount.
$ab,\ a(b),\ a \cdot b,\ a \times b$	Each means multiply a by b. For example, if $a = 3$ and $b = 4$, then $ab = 12$. Writing letters next to each other does not join their digits.
$a/b,\ \dfrac{a}{b},\ a \div b$	Divide a by b. In a fraction, a is the numerator and b the denominator. The denominator cannot be zero. Example: $6/4 = 3/2$.
$(\,),\ [\,]$	Group operations to be performed together. A fraction bar also groups its entire numerator and denominator. Thus $1/[k(k+2)]$ divides 1 by the whole product $k(k+2)$.
$=,\ \neq,\ \approx$	Equal to; not equal to; approximately equal to. Use $\approx$ for a rounded answer: $1/3 \approx 0.333$, but $1/3 \neq 0.333$ exactly.
$<,\ >,\ \leq$	Less than; greater than; less than or equal to. For example, $2 < 5$. The condition $1 \leq x < 10$ means that x is at least 1 and less than 10: 1 is included, but 10 is excluded.
a^n	A power: the raised number is the exponent. For positive integer n, multiply n copies of a. Example: $2^3 = 2 \cdot 2 \cdot 2 = 8$. For $a \neq 0$, $a^0 = 1$.
a^{-n}	A negative exponent means a reciprocal, not a negative answer: $a^{-n} = 1/a^n$ for $a \neq 0$. Thus $10^{-3} = 1/1000 = 0.001$.
$\sqrt{x}$	The nonnegative square root: the nonnegative number whose square is x. For example, $\sqrt{9} = 3$. The equation $y^2 = 9$ has two solutions, 3 and -3, but $\sqrt{9}$ itself is 3.
$\lvert x \rvert$	Absolute value: distance from zero, always nonnegative. For example, $\lvert -4 \rvert = 4$ and $\lvert 4 \rvert = 4$. The bars do not mean parentheses.
$V_1,\ V_2,\ x_i$	Subscripts label quantities or items; they are not exponents. V_1 and V_2 can label two volumes. x_i means the value with label i, so x_3 is the third value.

Notation	Meaning and example (continued)
$\bar{x},\ v_{\text{avg}}$	The bar in $\bar{x}$ denotes an average. The word avg in v_{avg} labels average speed. Example: the average of $2, 4, 9$ is $\bar{x} = (2 + 4 + 9)/3 = 5$.
$f(k)$	The value of an expression or function at input k. If $f(k) = 1/[k(k+1)]$, then $f(2) = 1/6$. Here $f(k)$ is function notation, not f multiplied by k.
$\sum_{k=1}^{n} f(k)$	Add the terms from $k = 1$ through $k = n$, including both endpoints. The letter k is the index. For example, $\sum_{k=1}^{3} k^2 = 1^2 + 2^2 + 3^2 = 14$.
$\prod_{k=2}^{n} f(k)$	Multiply the terms from $k = 2$ through $k = n$. For example, $\prod_{k=2}^{4} k = 2 \cdot 3 \cdot 4 = 24$. The large symbol is a product sign, not the number π.
$w_i x_i,\ \sum w_i$	w_i is the weight of value x_i, and $w_i x_i$ is their product. $\sum w_i$ adds all weights in the problem. For class averages, a weight can be the number of students in a class.
$\cdots$	Continue the displayed pattern. For example, $1 + 2 + \cdots + 5$ means $1 + 2 + 3 + 4 + 5$. The surrounding terms specify which pattern to continue.
$\max(u, v)$	Choose the larger value. For example, $\max(2, 4) = 4$. If the values are equal, their common value is the maximum.
$\Longrightarrow$	Implies, or therefore: the statement on the left leads to the statement on the right. For example, $2x = 6 \Longrightarrow x = 3$.
If and only if	A condition works in both directions: each statement guarantees the other. For integers, a number is even if and only if it is divisible by 2.
$a : b,\ p\%$	$a : b$ is the ratio of a to b. A ratio $2 : 3$ has 5 total parts. $p\%$ means $p/100$; for example, $15\% = 0.15$.
cm^2, cm^3, kg/m^3	Square centimeters measure area; cubic centimeters measure volume. Kilograms per cubic meter measure density. Powers on units matter: $1\ \text{m}^2 = 10{,}000\ \text{cm}^2$.
L, h, s, mph	Liters, hours, seconds, and miles per hour. A slash in units means per: 20 m/s means 20 meters each second.

Notation	Meaning and example (continued)
P, I, A	In interest problems, P is the principal (original amount invested), I is interest earned, and A is the final balance, including principal and interest. In purchase problems, P is the original price and A is the final amount paid.
d, q, r, t	In the money section, d is the discount rate, q the tax rate, and r the annual interest rate, all written as decimals. t is time in years; for example, 9 months gives $t = 3/4$.
1.3.2; C1	Exercise 1.3.2 is exercise 2 in Section 1.3. C1 is challenge problem 1. Each solution has the same label as its problem.

1.1 Numbers, decimals, and efficient comparison

Formula. For positive denominators,

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}.$$

A reduced fraction has a terminating decimal if and only if its denominator is $2^u 5^v$, where u, v are nonnegative integers. Its minimum number of decimal places is $\max(u, v)$.

Explanation. For close fractions, compare cross-products or their distances from a convenient benchmark. Do not convert to rounded decimals when exact comparison is easy. Negative signs reverse the order: if $x > y$, then $-x < -y$.

The decimal rule follows from place value: a terminating decimal has denominator $10^n = 2^n 5^n$. Reduce first; factors that cancel do not prevent termination.

Example. Which is larger, 17/29 or 11/19? Since

$$17 \cdot 19 = 323 > 319 = 11 \cdot 29,$$

17/29 is larger. Their exact difference is 4/551.

Harder example. How many decimal places are needed to write 117/7500 exactly? Reduce:

$$\frac{117}{7500} = \frac{39}{2500}, \qquad 2500 = 2^2 5^4.$$

Four places are needed. Multiplying numerator and denominator by 4 gives $156/10000 = 0.0156$.

Practice.

1.1.1. Which is greater: $-17/29$ or $-11/19$?

1.1.2. Which is larger, 1001/1002 or 1002/1003? Find their difference.

1.1.3. Find the least positive integer n such that $n/840$ has a terminating decimal.

Solutions.

1.1.1. $-11/19$ is greater, because $11/19 < 17/29$ and negation reverses the inequality.

1.1.2. Write the fractions as $1 - 1/1002$ and $1 - 1/1003$. The second is larger, by

$$\frac{1}{1002} - \frac{1}{1003} = \frac{1}{1{,}005{,}006}.$$

1.1.3. $840 = 2^3 \cdot 3 \cdot 5 \cdot 7$. The numerator must cancel $3 \cdot 7$, so the least choice is $n = 21$. Indeed, $21/840 = 1/40$.

1.2 Order of operations and mental structure

Formula.

$$(a - b)(a + b) = a^2 - b^2, \qquad (a + b)^2 = a^2 + 2ab + b^2.$$

For a nonnegative integer n,

$$(10n + 5)^2 = 100n(n + 1) + 25.$$

Explanation. Use parentheses and exponents first, then multiplication and division from left to right, then addition and subtraction from left to right. A fraction bar groups its entire numerator and denominator.

Look for a convenient center before multiplying. The difference of squares is especially useful for equally spaced factors. A square ending in 5 can be found by multiplying the preceding integer by its successor and appending 25.

Example.

$$97 \cdot 103 = (100 - 3)(100 + 3) = 10{,}000 - 9 = 9991.$$

Similarly, 85^2 uses $8 \cdot 9 = 72$, giving 7225.

Harder example. Evaluate $(1003^2 - 997^2)/(6 \cdot 125)$. Factor before computing squares:

$$\frac{(1003 - 997)(1003 + 997)}{6 \cdot 125} = \frac{6 \cdot 2000}{6 \cdot 125} = 16.$$

Practice.

1.2.1. Evaluate $-3^2 + (-3)^2 + 24 \div 6 \cdot 2$.

1.2.2. Compute 995^2 mentally.

1.2.3. Evaluate $(2027^2 - 2023^2)/(2027 - 2023)$.

Solutions.

1.2.1. $-9 + 9 + 4 \cdot 2 = 8$. The exponent in -3^2 applies to 3, not to the minus sign.

1.2.2. $(1000 - 5)^2 = 1{,}000{,}000 - 10{,}000 + 25 = 990{,}025$.

1.2.3. Factor the numerator and cancel the difference. The result is $2027 + 2023 = 4050$.

1.3 Equivalent fractions, cancellation, and telescoping

Formula. For nonzero denominators and $c \neq 0$,

$$\frac{a}{b} = \frac{ac}{bc}, \qquad \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Explanation. Reduce products before multiplying. Only factors cancel: $(a+b)/(a+c)$ does not simplify by removing a. In a long sum or product, write the first few terms and the last term to see what survives.

Example.

$$\frac{84 \cdot 125}{35 \cdot 72} = \frac{7}{6} \cdot \frac{25}{7} = \frac{25}{6}.$$

Harder example. Evaluate $\prod_{k=2}^{10}(1 - 1/k^2)$. Factor each term:

$$\prod_{k=2}^{10} \frac{(k-1)(k+1)}{k^2} = \left(\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{9}{10}\right)\left(\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{11}{10}\right) = \frac{1}{10} \cdot \frac{11}{2} = \frac{11}{20}.$$

Here $\prod$ means multiply all terms in the indicated range.

Practice.

1.3.1. Reduce 462/1078.

1.3.2. Evaluate $\displaystyle\sum_{k=1}^{19} \frac{1}{k(k+1)}$.

1.3.3. Evaluate $\displaystyle\sum_{k=1}^{10} \frac{1}{(2k-1)(2k+1)}$.

Solutions.

1.3.1. Divide numerator and denominator by 154: the result is 3/7.

1.3.2. Each term is $1/k - 1/(k+1)$, so the sum is $1 - 1/20 = 19/20$.

1.3.3. Use

$$\frac{1}{(2k-1)(2k+1)} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right).$$

The sum is $\frac{1}{2}(1 - 1/21) = 10/21$.

1.4 Ratios, proportions, and scale factors

Formula. If $x : y = a : b$, write $x = at$ and $y = bt$. For similar figures with positive length scale factor s,

$$\frac{A_2}{A_1} = s^2, \qquad \frac{V_2}{V_1} = s^3.$$

Explanation. Ratio parts become actual quantities only after finding the common multiplier. When joining two ratios, make the shared quantity have the same number of parts. For map areas or model volumes, square or cube the scale factor before converting units.

Example. If $x : y = 3 : 5$ and $y : z = 10 : 7$, rewrite the first ratio as $6 : 10$. Then

$$x : y : z = 6 : 10 : 7.$$

If $x + y + z = 92$, one part is $92/23 = 4$, so $(x, y, z) = (24, 40, 28)$.

Harder example. A map has scale $1 : 50{,}000$. A region has map area 12 cm^2. One centimeter represents 0.5 km, so

$$A = 12(0.5)^2 = 3 \text{ km}^2.$$

Using the linear factor just once would give the wrong area.

Practice.

1.4.1. If $x : y = 7 : 11$ and $y - x = 36$, find $x + y$.

1.4.2. Similar solids have volumes in ratio $27 : 125$. Find their surface-area ratio.

1.4.3. A mixture has red and blue beads in ratio $3 : 5$. Adding 12 red beads changes the ratio to $3 : 4$. How many beads were there originally?

Solutions.

1.4.1. Four parts equal 36, so a part is 9. The sum is $18(9) = 162$.

1.4.2. The length ratio is $3 : 5$, so the area ratio is $9 : 25$.

1.4.3. Let the original counts be $3t$ and $5t$.

$$\frac{3t + 12}{5t} = \frac{3}{4} \quad \Longrightarrow \quad 12t + 48 = 15t \quad \Longrightarrow \quad t = 16.$$

There were $8t = 128$ beads.

1.5 Percent increase, decrease, and reversal

Formula. Write a percentage as a decimal r. An increase multiplies by $1 + r$; a decrease multiplies by $1 - r$.

$$\text{fractional change} = \frac{\text{new} - \text{old}}{\text{old}}.$$

A decrease of fraction $r < 1$ requires a fractional increase $r/(1 - r)$ to reverse it.

Explanation. Successive changes multiply because each uses the current value as its base. A change of base also explains why a 20% decrease takes a 25% increase to reverse. Percentage points refer to subtraction of percentages: 30% to 40% is 10 percentage points, but a $33\frac{1}{3}\%$ relative increase.

Example. An increase of 20% followed by a decrease of 20% gives

$$(1.2)(0.8) = 0.96.$$

The net result is a 4% decrease.

Harder example. A rectangle's length increases by 25%. By what percentage must its width decrease to preserve area? The width multiplier must satisfy

$$1.25w = 1, \qquad w = 0.8.$$

The width must decrease by 20%.

Practice.

1.5.1. A population grows by 10% each year for two years. Find its total percentage increase.

1.5.2. A price falls by 15%. Find the exact percentage increase needed to restore it.

1.5.3. A quantity is increased twice by the same percentage and becomes 144% of its original value. Find that percentage.

Solutions.

1.5.1. $1.1^2 = 1.21$, so the increase is 21%.

1.5.2. The increase is $(15/85)100\% = 300/17\% \approx 17.647\%$.

1.5.3. $(1 + r)^2 = 1.44$, so $1 + r = 1.2$ and $r = 0.2$. Each increase was 20%.

1.6 Unit conversions and controlled estimation

Formula.

$$1 \text{ m}^2 = 10^4 \text{ cm}^2, \qquad 1 \text{ m}^3 = 10^6 \text{ cm}^3, \qquad 1 \text{ km/h} = \frac{5}{18} \text{ m/s}.$$

Explanation. Conversion factors equal 1; orient them so units cancel. An area or volume conversion uses the square or cube of a length conversion. For estimates, replace numbers by nearby convenient values, then decide whether the estimate is above or below the true value. Do not treat an estimate as an exact answer.

Example. Convert 72 km/h:

$$72\frac{\text{km}}{\text{h}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ h}}{3600 \text{ s}} = 20 \text{ m/s}.$$

Harder example. Convert 1.2 g/cm^3 to kg/m^3:

$$1.2\frac{\text{g}}{\text{cm}^3} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{10^6 \text{ cm}^3}{1 \text{ m}^3} = 1200 \text{ kg/m}^3.$$

The denominator conversion makes the number larger.

Practice.

1.6.1. Convert 0.008 m^3 to cm^3.

1.6.2. Water flows at 250 cm^3/s. How many liters flow in 6 minutes?

1.6.3. Without multiplying out, determine whether $198 \cdot 51$ is above or below 10,000, and find its exact difference from 10,000.

Solutions.

1.6.1. $0.008(10^6) = 8000$ cm^3.

1.6.2. 250 cm^3/s is 0.25 L/s. In 360 s, the volume is $0.25(360) = 90$ L.

1.6.3. $(200-2)(50+1) = 10{,}000 + 200 - 100 - 2$. The product is 98 above 10,000, namely 10,098.

1.7 Rates, work, and weighted averages

Formula.

$$\bar{x} = \frac{\sum w_i x_i}{\sum w_i}, \qquad v_{\text{avg}} = \frac{\text{total distance}}{\text{total time}}.$$

For independent workers with constant rates and no interference,

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b}.$$

For two equal-distance legs at positive speeds u, v, the average speed is $2uv/(u+v)$.

Explanation. Find the conserved total: score totals for class averages, distance and time for speed, and job fractions for work. Add rates, never completion times. Equal-time speed legs have equal time weights; equal-distance legs generally do not.

Example. Twenty students average 84, and thirty average 90. The combined average is

$$\frac{20(84) + 30(90)}{50} = 87.6.$$

Harder example. A car travels one-third of a trip at 30 mph and the rest at 60 mph. Choose a total distance of 90 miles. Time is $30/30 + 60/60 = 2$ h, so average speed is $90/2 = 45$ mph.

Practice.

1.7.1. Two machines finish a job in 6 h and 3 h individually. Find their combined completion time.

1.7.2. A pipe fills a tank in 4 h and a drain empties it in 6 h. Starting empty, how long does filling take with both open? Assume constant rates.

1.7.3. A car travels half the distance at 40 mph. What speed on the remaining half makes its overall average 48 mph?

Solutions.

1.7.1. Rate $= 1/6 + 1/3 = 1/2$ job/h, so time is 2 h.

1.7.2. Net rate $= 1/4 - 1/6 = 1/12$ tank/h, so time is 12 h.

1.7.3. Let each half be 120 miles. Total time must be $240/48 = 5$ h. The first half takes 3 h, leaving 2 h; the required speed is $120/2 = 60$ mph.

1.8 Money, discounts, and interest

Formula. For a purchase with a discount followed by tax,

$$A = P(1 - d)(1 + q).$$

Here P is the original price, A is the final amount paid, d is the discount rate, and q is the tax rate. Write both rates as decimals.

For simple interest,

$$I = Prt, \qquad A_{\text{simple}} = P + I = P(1 + rt).$$

Here P is the principal (the original amount invested), I is the interest earned, A is the final balance, r is the annual interest rate as a decimal, and t is time in years. With annual compounding at the same annual rate,

$$A_{\text{compound}} = P(1 + r)^t$$

for an integer number of years.

Explanation. Identify the base of each percentage. Tax on a discounted price uses the sale price. Simple interest uses only the original principal; compound interest uses the growing balance. Keep exact values until rounding money at the end, unless the question specifies another convention.

Example. A \$120 item receives a 25% discount and then 10% tax:

$$120(0.75)(1.10) = \$99.$$

Harder example. Compare simple and annually compounded interest on \$1000 at 10% for two years. The balances are \$1200 and \$1210. The extra \$10 comes from interest on the first year's interest:

$$P(1 + r)^2 - P(1 + 2r) = Pr^2.$$

Practice.

1.8.1. Find the simple interest on \$800 at 6% per year for 9 months.

1.8.2. An item's price after a 20% discount and 8% tax is \$172.80. Find its original price.

1.8.3. A store marks an item up by 40% above cost. What discount from that marked price leaves a 12% profit on cost?

Solutions.

1.8.1. $t = 9/12 = 3/4$, so $I = 800(0.06)(3/4) = \$36$.

1.8.2. $0.8(1.08)P = 172.80$, so $P = 172.80/0.864 = \$200$.

1.8.3. For cost C, require $1.4C(1-d) = 1.12C$. Then $1-d = 0.8$, so the discount is 20%.

1.9 Mixtures and replacement

Formula. If volumes are additive, mixing volumes V_1, V_2 with concentrations c_1, c_2 gives

$$c = \frac{c_1V_1 + c_2V_2}{V_1 + V_2}.$$

For a fixed-volume, well-mixed tank, replacing fraction f by pure water n times leaves fraction $(1-f)^n$ of the original solute.

Explanation. Track the solute amount rather than the concentration alone. Removing a well-mixed portion removes the same fraction of solute and solution; adding water restores volume but adds no solute. In ordinary mixture exercises, assume concentrations and volumes use compatible units and volumes add.

Example. Mix 6 L of 20% solution with 4 L of 50% solution:

$$c = \frac{6(0.20) + 4(0.50)}{10} = 0.32 = 32\%.$$

Harder example. A 10 L tank contains 30% solution. Remove 2 L and replace it with water; repeat once. Each replacement retains 8/10 of the solute, so the final concentration is

$$30\%(0.8)^2 = 19.2\%.$$

Practice.

1.9.1. How much pure water must be added to 6 L of 40% solution to obtain 24% solution?

1.9.2. How many liters of 40% solution must be mixed with 10% solution to make 18 L of 25% solution?

1.9.3. A 12 L tank holds 25% solution. Remove 3 L and replace it with 50% solution. Find the new concentration.

Solutions.

1.9.1. Solute amount is $6(0.4) = 2.4$ L. Final volume must be $2.4/0.24 = 10$ L, so add 4 L.

1.9.2. $0.4x + 0.1(18-x) = 0.25(18)$ gives $0.3x = 2.7$, so use 9 L of each.

1.9.3. The remaining 9 L contains 2.25 L of solute. The added solution supplies 1.5 L. Concentration is $3.75/12 = 31.25\%$.

1.10 Scientific notation and powers of ten

Formula. For nonzero a, b,

$$(a10^m)(b10^n) = ab10^{m+n}, \qquad \frac{a10^m}{b10^n} = \frac{a}{b}10^{m-n}, \qquad (a10^m)^k = a^k 10^{mk}.$$

Normalized scientific notation uses $1 \le |a| < 10$.

Explanation. Separate coefficients from exponents, then normalize. In sums, first give all terms the same exponent. For roots, rewrite the exponent as a multiple of the root's degree. Check the scale before calculating the coefficient.

Example.

$$\frac{6.3 \times 10^{12}}{2.1 \times 10^5} = 3 \times 10^7.$$

Harder example.

$$\sqrt{6.4 \times 10^{-7}} = \sqrt{64 \times 10^{-8}} = 8 \times 10^{-4}.$$

Rewriting the coefficient made the exponent even.

Practice.

1.10.1. Compute $(8 \times 10^{-3})(2.5 \times 10^6)$.

1.10.2. Write $3.2 \times 10^{-4} + 7.5 \times 10^{-5}$ in normalized scientific notation.

1.10.3. Evaluate $\dfrac{(3 \times 10^{-4})^2}{6 \times 10^{-11}}$.

Solutions.

1.10.1. $20 \times 10^3 = 2 \times 10^4$.

1.10.2. $(3.2 + 0.75)10^{-4} = 3.95 \times 10^{-4}$.

1.10.3. $(9 \times 10^{-8})/(6 \times 10^{-11}) = 1.5 \times 10^3$.

Challenge problems

C1. Find the least positive integer n such that both $n/1260$ and $n/924$ have terminating decimals.

C2. Evaluate $\dfrac{10001^2 - 9999^2}{0.125}$.

C3. Evaluate $\displaystyle\sum_{k=1}^{20} \frac{1}{k(k+2)}$.

C4. Evaluate $\displaystyle\prod_{k=2}^{50} \left(1 - \frac{1}{k^2}\right)$.

C5. A mixture has red and blue beads in ratio $2 : 3$. After adding 10 red beads and removing 5 blue beads, the ratio is $3 : 4$. Find the original total.

C6. Two similar containers have volumes in ratio $8 : 27$. The larger has surface area 243 cm^2. Find the smaller's surface area.

C7. A price rises by $p\%$ and then falls by $p\%$. The final price is 91% of the original. Find p.

C8. A car travels 40% of a trip at 30 mph, 40% at 60 mph, and the remainder at 90 mph. Find its exact average speed.

C9. A and B finish a job in 6 h together; B and C in 8 h; A and C in 12 h. With constant independent rates, how long do all three need?

C10. On the same principal and annual rate, two-year compound interest exceeds two-year simple interest by \$45. The first year's interest is \$300. Find the principal and rate.

C11. A 20 L tank contains 40% solution. Remove 5 L and replace with water; mix and repeat. How much pure solute must then be added to make a 30% solution? Assume added volumes are additive.

C12. A substance has density 2.4 g/cm^3. Its mass is 3.6 kg. Find its volume in m^3, in scientific notation.

Challenge solutions

C1. $1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7$ and $924 = 2^2 \cdot 3 \cdot 7 \cdot 11$. To leave only factors 2 and 5 in both reduced denominators, n must contain 3^2, 7, and 11. Thus the least choice is

$$n = 3^2 \cdot 7 \cdot 11 = 693.$$

Indeed, $693/1260 = 11/20$ and $693/924 = 3/4$.

C2. Factor the numerator:

$$\frac{(10001 - 9999)(10001 + 9999)}{1/8} = 2(20000)(8) = 320{,}000.$$

C3. Use $1/[k(k+2)] = \frac{1}{2}(1/k - 1/(k+2))$. All but the first two and last two terms cancel:

$$\frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{21} - \frac{1}{22}\right) = \frac{3}{4} - \frac{43}{924} = \frac{325}{462}.$$

C4. Separate the factors:

$$\left(\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{49}{50}\right)\left(\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{51}{50}\right) = \frac{1}{50} \cdot \frac{51}{2} = \frac{51}{100}.$$

C5. Write the original counts as $2t, 3t$. Then

$$\frac{2t + 10}{3t - 5} = \frac{3}{4} \quad \Longrightarrow \quad 8t + 40 = 9t - 15 \quad \Longrightarrow \quad t = 55.$$

The original total was $5t = 275$ beads.

C6. The length ratio is $2 : 3$ and the area ratio is $4 : 9$. The smaller area is $243(4/9) = 108$ cm^2.

C7. $(1 + p/100)(1 - p/100) = 0.91$. Thus $(p/100)^2 = 0.09$. Since p is positive, $p = 30$.

C8. Choose a 450 mile trip. The distances are $180, 180, 90$ miles, with times $6, 3, 1$ h. The average is $450/10 = 45$ mph.

C9. If individual rates are a, b, c, then

$$a + b = \frac{1}{6}, \quad b + c = \frac{1}{8}, \quad a + c = \frac{1}{12}.$$

Adding gives $2(a + b + c) = 3/8$, so their combined rate is $3/16$ job/h. Time is $16/3$ h, or 5 h 20 min.

C10. The excess is $Pr^2 = 45$, while first-year interest is $Pr = 300$. Dividing gives $r = 45/300 = 0.15$, so the annual rate is 15% and $P = 300/0.15 = \$2000$.

C11. After replacement, solute volume is

$$20(0.4)\left(\frac{15}{20}\right)^2 = 4.5 \text{ L}.$$

Adding x L of pure solute changes both solute and total volume:

$$\frac{4.5 + x}{20 + x} = 0.30 \quad \Longrightarrow \quad 0.7x = 1.5.$$

Add $x = 15/7$ L. Adding only 1.5 L would overlook the increase in total volume.

C12. 2.4 g/cm^3 = 2400 kg/m^3. Therefore

$$V = \frac{3.6 \text{ kg}}{2400 \text{ kg/m}^3} = 1.5 \times 10^{-3} \text{ m}^3.$$